Active Directory Migration Lab

Windows Server 2008 R2 to Windows Server 2022 | Cross-forest directory-object migration
Source: PADA-AD / wfr-fr.net (public aliases) | Destination: CASTILLUX-DC01 / ad.castillux.com
Technical case study.

Executive summary

This isolated Oracle VirtualBox laboratory recreated directory objects from a legacy Windows Server 2008 R2 forest in a separate Windows Server 2022 forest. The legacy domain was preserved rather than upgraded in place. The work included organisational units, security groups, users, group memberships, computer accounts and selected Group Policy Objects. Source identifiers in this public-facing edition are anonymised.

Deliverable	Observed result
Custom organisational units	20 recreated
Security groups	21 recreated (19 global, 2 domain-local)
User accounts	129 recreated; disabled
Direct user-to-group memberships	153 restored and verified
Legacy computer objects	223 inventoried; 222 recreated as disabled accounts
Historical Group Policy Objects	38 backed up; 10 LAB- GPOs imported and left unlinked
Domain controller validation	DCDIAG connectivity, advertising, services and DNS passed
Shares	SYSVOL and NETLOGON published

1. Scope and safety boundaries

The destination forest has different security identifiers. This is an object recreation exercise, not a SID-preserving or password-preserving migration. No forest trust, password migration, physical client domain join, production cutover or decommissioning of the source domain was performed. The imported user and computer accounts were disabled. Legacy printers and NAS shares were not migrated.

2. Architecture and addressing

System	Role	Lab network
PADA-AD	Windows Server 2008 R2; source AD DS	NAT 10.0.2.15; isolated 172.20.10.10/24
CASTILLUX-DC01	Windows Server 2022; destination AD DS and DNS	Bridged 192.168.1.10/24; isolated 172.20.10.20/24
HP Z400	Ubuntu Linux VirtualBox host	Virtualisation host

The isolated AD-MIGRATION adapters used no default gateway and were not dynamically registered in DNS. A destination share was mapped as drive Z: on the source for exports. Conditional DNS forwarders supported name resolution during the exercise.

Deployment evidence

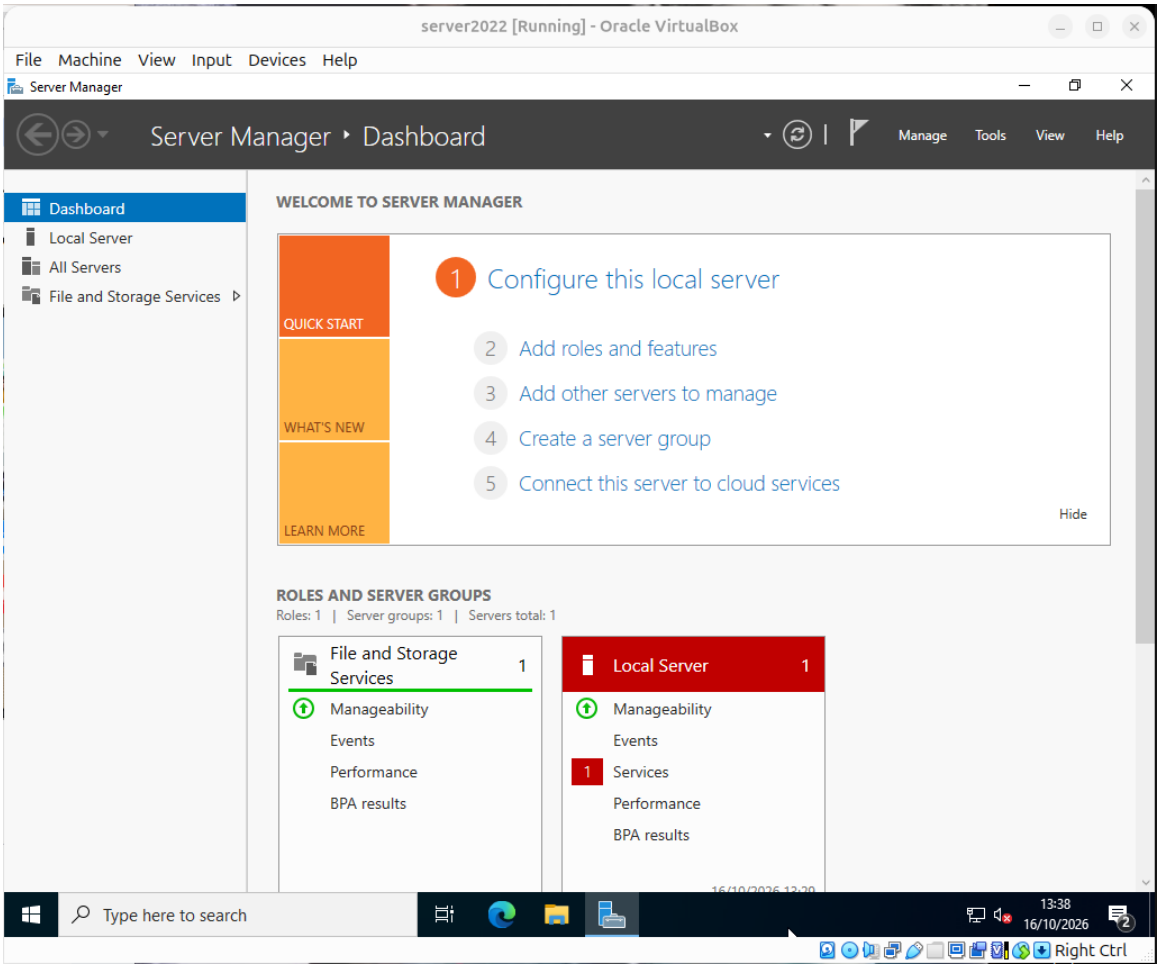


Figure 1 — Initial laboratory configuration (screenshot ad-01).

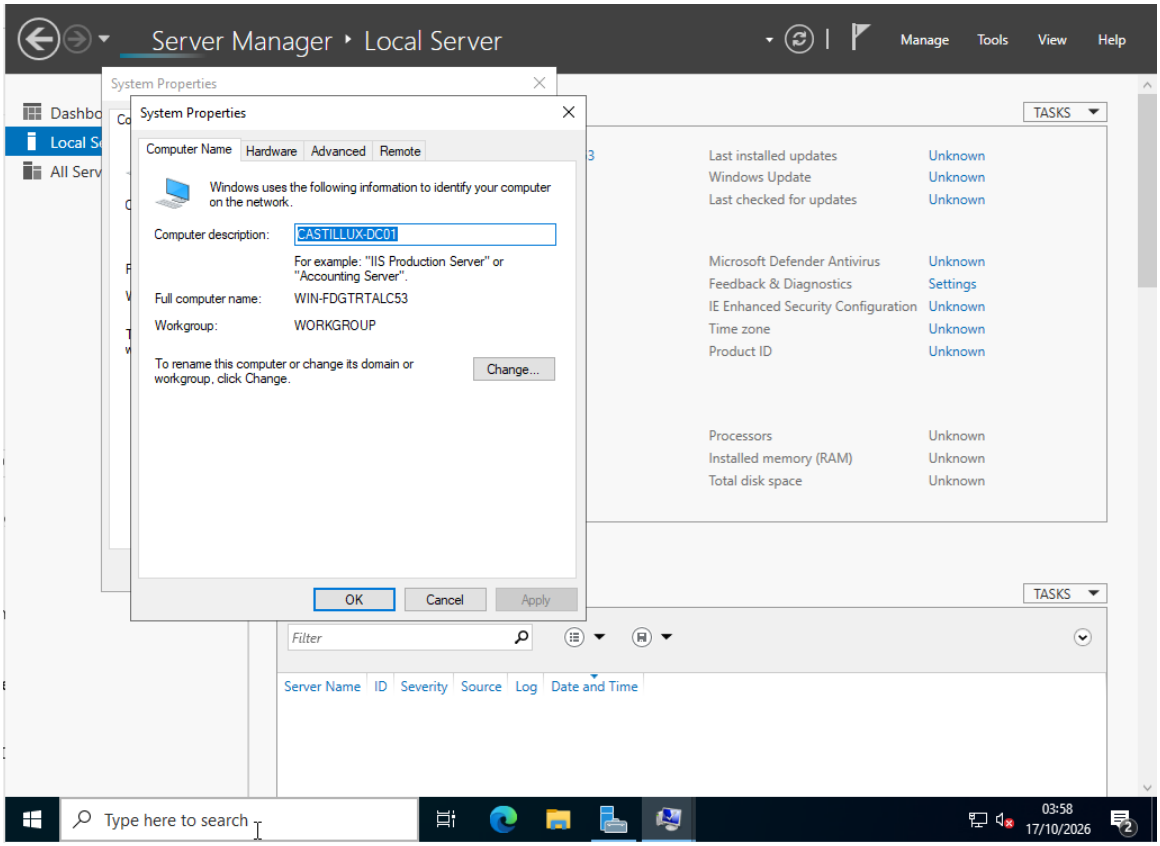


Figure 2 — Windows Server configuration (screenshot ad-03).

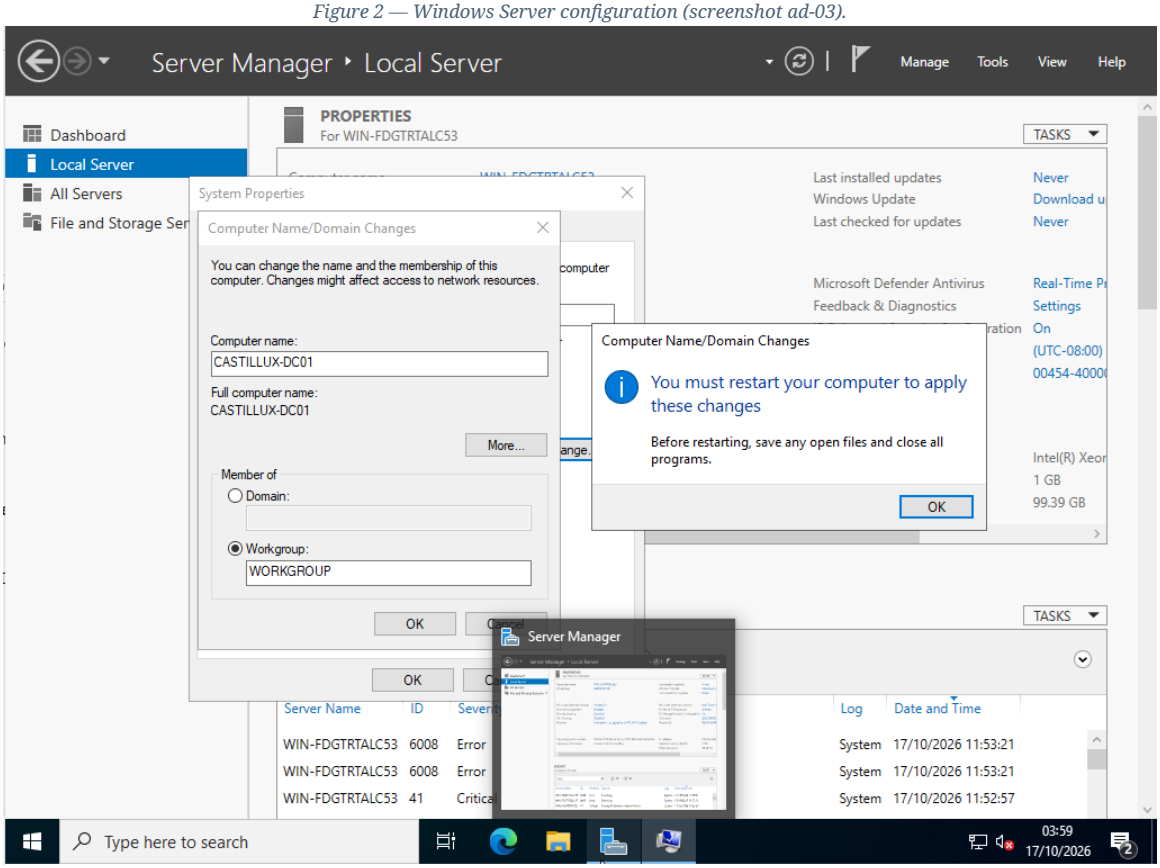


Figure 3 — Infrastructure preparation (screenshot ad-04).

Networking evidence

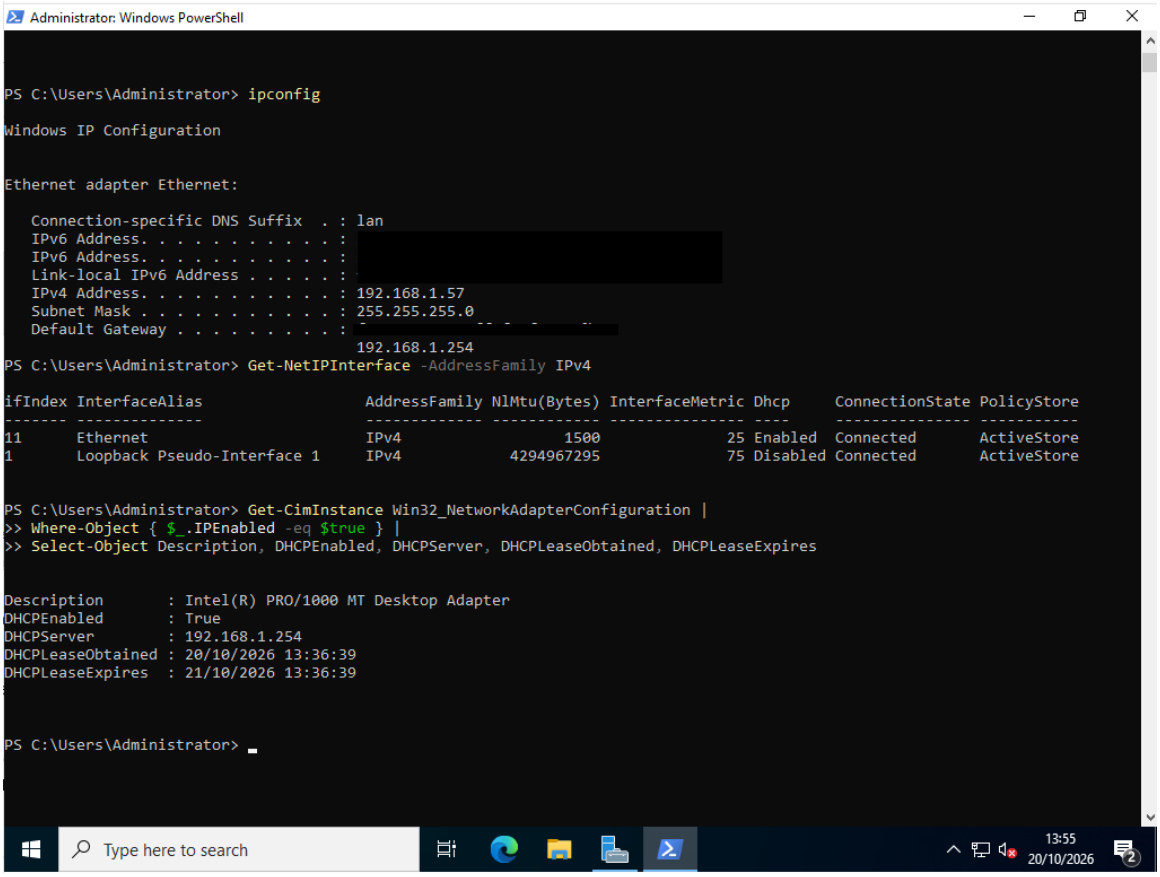


Figure 4 — Network configuration evidence (screenshot ad-15).

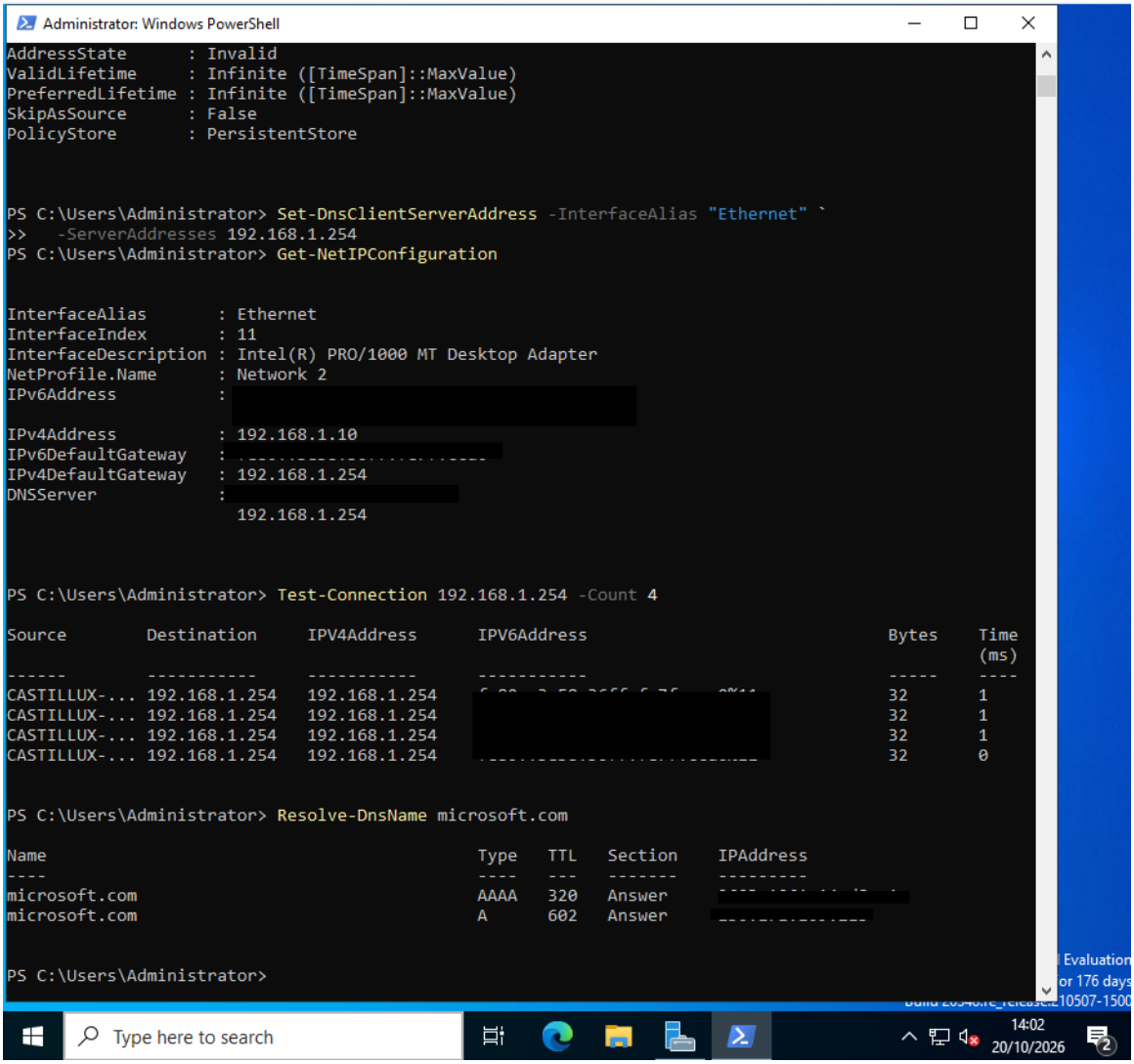


Figure 5 — Migration network preparation (screenshot ad-16).

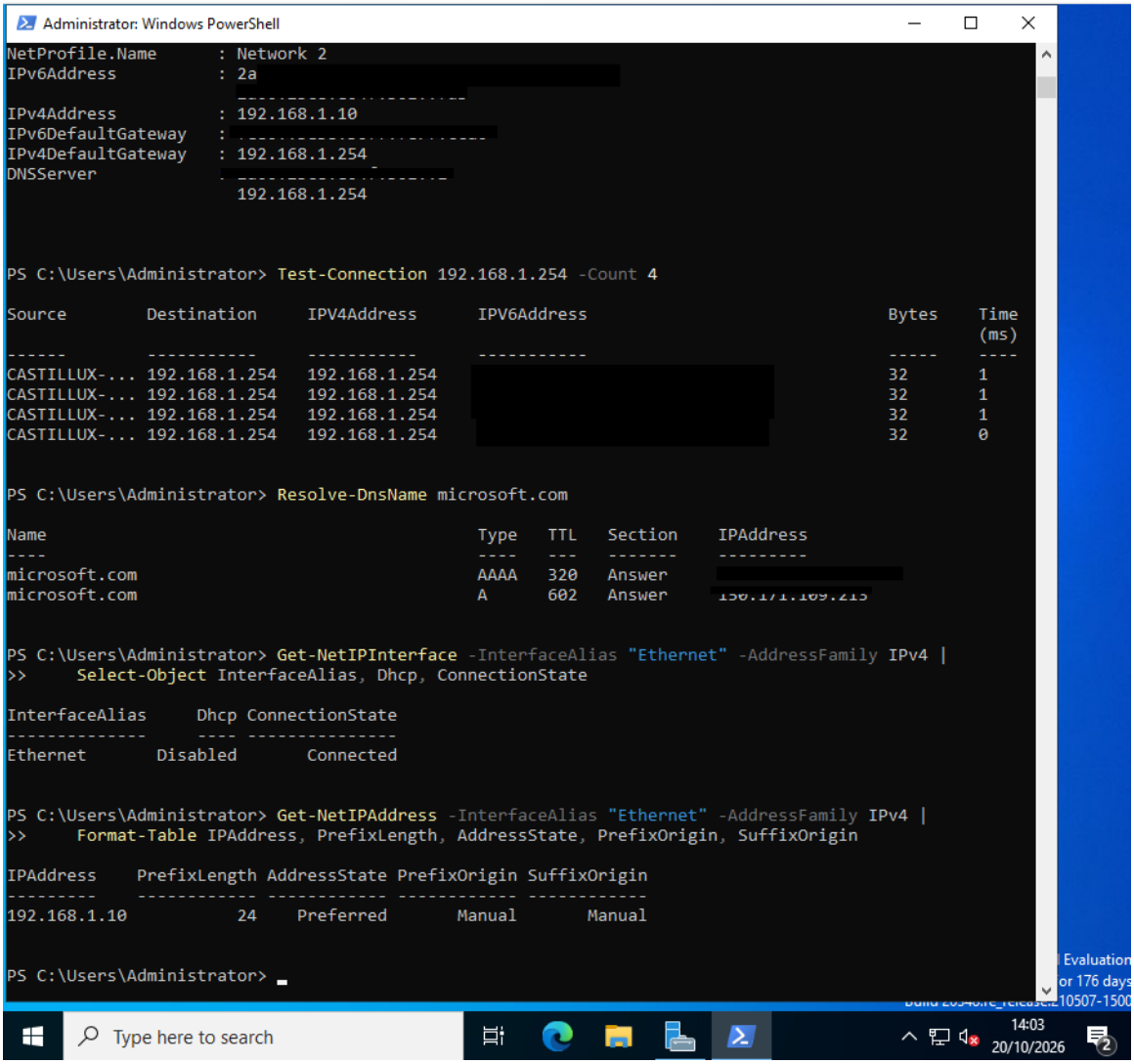


Figure 6 — Network connectivity configuration (screenshot ad-17).

3. Preparation and transfer

The preparation sequence included source inventory, destination forest deployment, isolated inter-VM networking, DNS checks, a transfer directory and a mapped share. PowerShell 2.0-compatible commands were required on the Windows Server 2008 R2 source.

```
New-Item -Path "C:\AD-Migration" -ItemType Directory -Force
```

The original export/import scripts for all organisational units, groups and users were not retained in the supplied technical record; their observed totals are documented without inventing commands.

4. Directory-object migration

4.1 Organisational units and groups

Twenty custom OUs were recreated in the target while preserving the legacy hierarchy. Twenty-one security groups were reconstructed (19 global and two domain-local). The target forest assigned new SIDs.

```
Import-Module ActiveDirectory
Get-ADGroup -Identity "Informatique" | Select-Object Name,GroupScope,GroupCategory
Get-ADGroup -Identity "Comptabilite" | Select-Object Name,GroupScope,GroupCategory
```

4.2 Users and memberships

A total of 129 migrated users were recreated in their destination containers and deliberately disabled. Passwords and original SIDs were not transferred. A later enumeration returned 132 accounts: the 129 migrated users plus Administrator, Guest and krbtgt. The technical record showed one enabled and 131 disabled accounts in that enumeration. The exercise restored and previously verified 153 direct group memberships; no nested memberships were found among the 21 selected groups.

```
$users = @(Get-ADUser -Filter * -Properties Enabled)
$enabled = @($users | Where-Object { $_.Enabled -eq $true })
$disabled = @($users | Where-Object { $_.Enabled -eq $false })
Write-Host "Total domain users:" $users.Count
Write-Host "Enabled:" $enabled.Count
Write-Host "Disabled:" $disabled.Count
```

4.3 Computer-account migration — completed

The source inventory contained 223 computer objects, including the legacy domain controller. The target received 222 disabled computer accounts, preserving their names and OU locations where applicable. One object was created as a test, followed by 221 objects in the corrected bulk import. The original source domain controller was intentionally excluded. This does not mean the physical machines were joined to the new domain.

```
# Source (PowerShell 2.0)
Import-Module ActiveDirectory
Get-ADComputer -Filter * -Properties OperatingSystem |
 Select-Object Name,SamAccountName,DistinguishedName,OperatingSystem |
 Export-Csv "Z:\Computers-Migration.csv" -NoTypeInformation -Encoding UTF8
```

The first bulk import returned Created: 0, Skipped: 1, Errors: 222. The Get-ADComputer -Identity existence check raised identity-not-found errors before New-ADComputer ran. Preloading existing names and comparing with -contains resolved the control-flow issue.

```
# Destination (illustrative corrected logic; source aliases substituted)
$computers = Import-Csv "C:\AD-Migration\Computers-Migration.csv"
$existingNames = @(Get-ADComputer -Filter * | Select-Object -ExpandProperty Name)
$created = 0; $skipped = 0; $errors = 0
foreach ($computer in $computers) {
 if ($computer.Name -eq "PADA-AD") { $skipped++; continue }
 if ($existingNames -contains $computer.Name) { $skipped++; continue }
 $parentDN = $computer.DistinguishedName -replace '^CN=[^,]+,', ''
 $targetDN = $parentDN.Replace("DC=wfr-fr,DC=net", "DC=ad,DC=castillux,DC=com")
 try {
   New-ADComputer -Name $computer.Name `
    -SamAccountName $computer.SamAccountName `
    -Path $targetDN -Enabled $false -ErrorAction Stop
  $created++; $existingNames += $computer.Name
 } catch { $errors++; Write-Host "ERROR:" $computer.Name $_.Exception.Message }
}
Write-Host "Created:" $created "Skipped:" $skipped "Errors:" $errors
```

Observed corrected run: Created 221; Skipped 2 (source DC and the test object); Errors 0. Including the earlier test object, 222 disabled source computer accounts were recreated. The final target total of 223 including CASTILLUX-DC01 was expected but not independently confirmed in the supplied record.

Migration implementation evidence

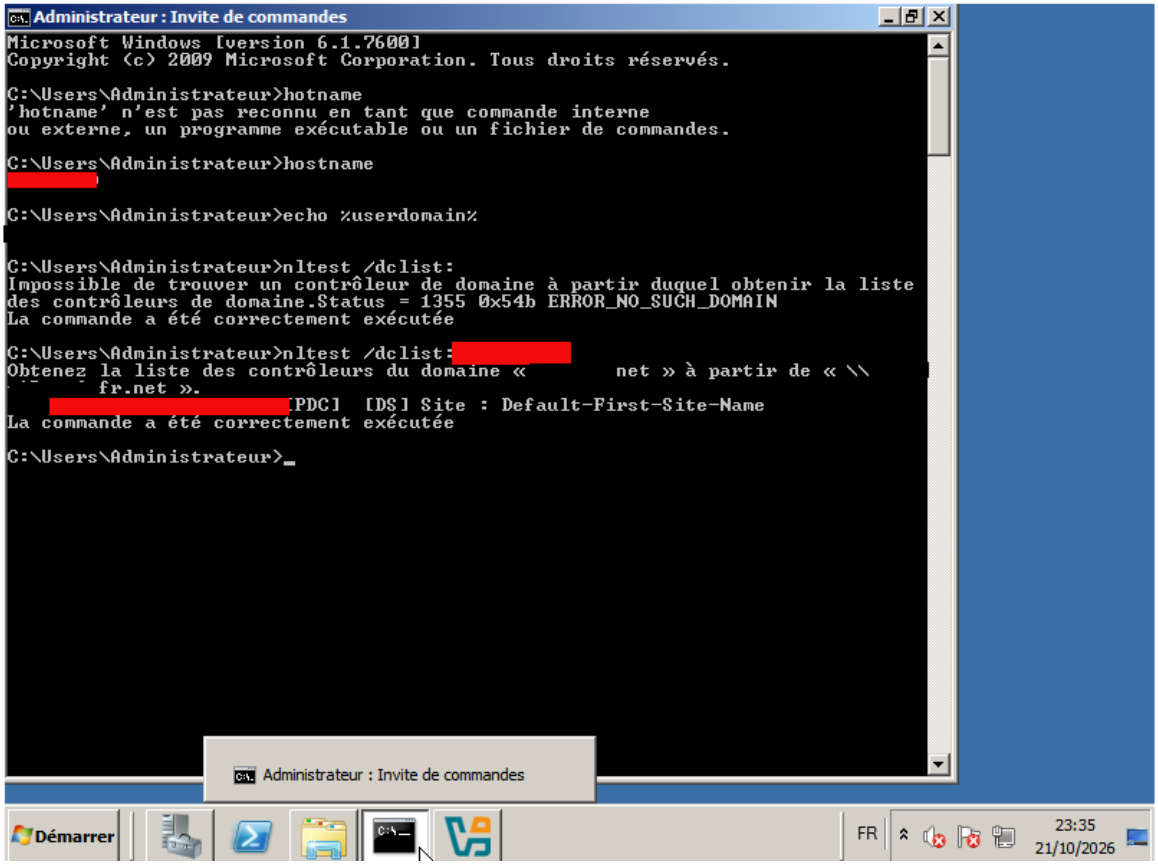


Figure 7 — Migration preparation (screenshot ad-36).

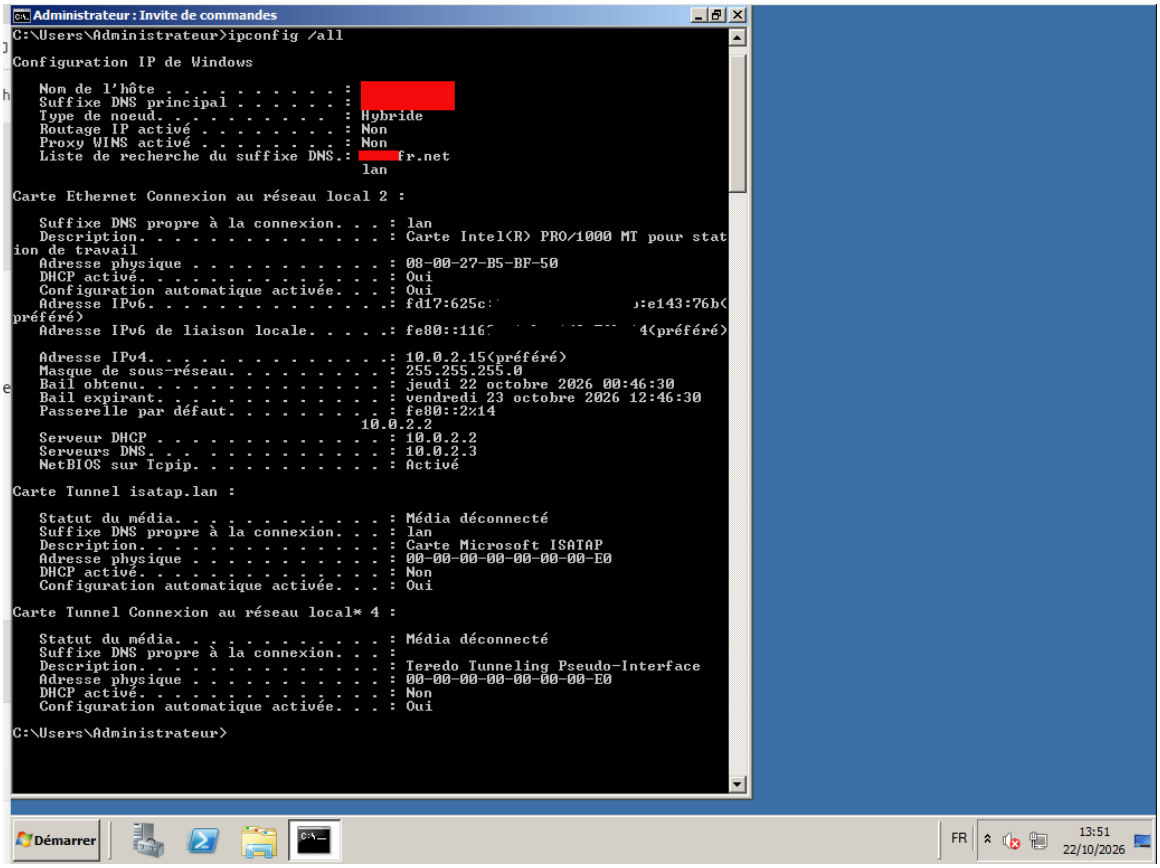


Figure 8 — Directory migration evidence (screenshot ad-39).

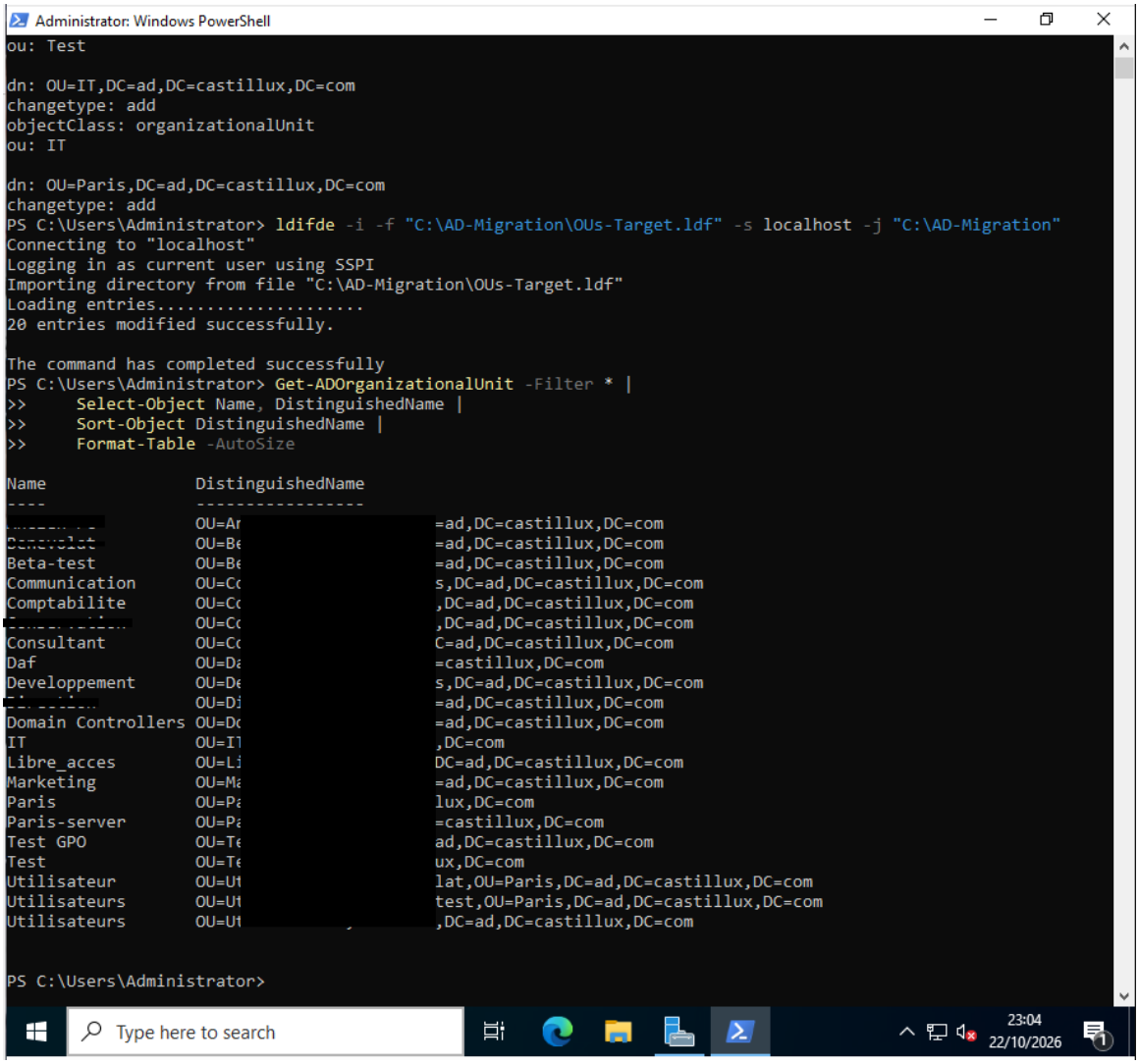


Figure 9 — Active Directory migration configuration (screenshot ad-42).

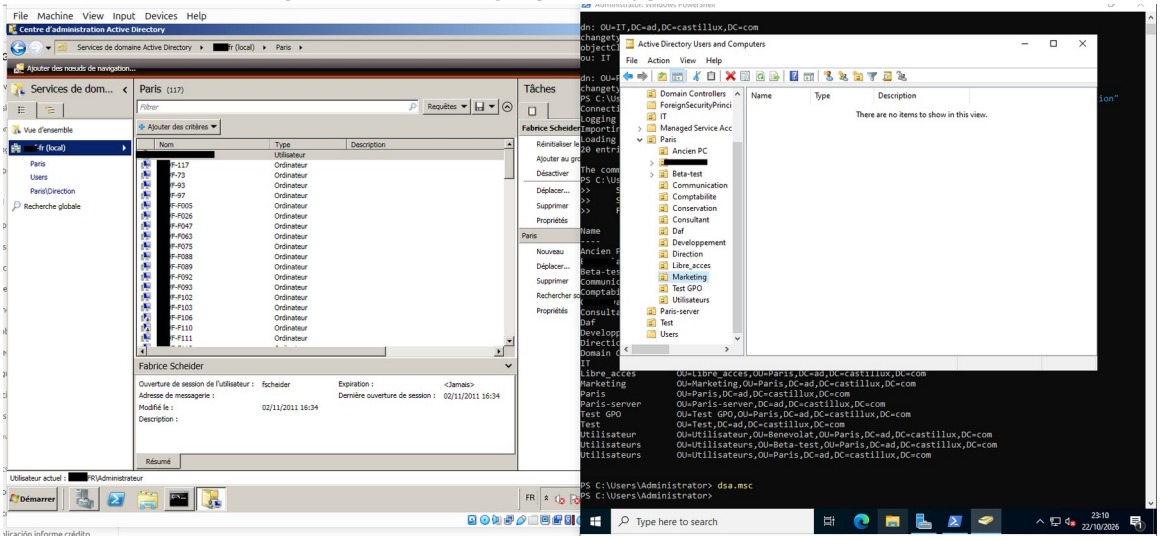


Figure 10 — Migration implementation evidence (screenshot ad-44).

5. Group Policy migration and assessment

The source held 38 distinct GPOs. Two backup runs produced 76 backup folders. The manifest and backup files were examined; eight selected policies plus LAB-Migration-Test and LAB-Remote-Desktop produced ten LAB-prefixed GPOs in the destination. They were left unlinked for safe review. The unavailable Get-GPOBackup cmdlet was replaced with the manifest-based workflow and Import-GPO.

```
# Source
Backup-GPO -All -Path "C:\AD-Migration\GPO-Backup"
# Destination
Import-Module GroupPolicy
Import-GPO -BackupGpoName "Test" -Path "C:\AD-Migration\GPO-Backup" `
 -TargetName "LAB-Migration-Test"
```

Legacy drive Q: mappings had overlapping targeting for Programmes, Comptabilite and Daf. Group membership overlap was reviewed and Daf precedence established. Remote Assistance authorisation was narrowed to CASTILLUX\Informatique. Printer connections and an obsolete startup script were identified; the script reference was removed, but printer migration was excluded. A historical firewall policy included obsolete exceptions and a VNC TCP 5900 rule; it was retained unlinked. The file-share audit policy report showed no configured user or computer settings.

Group Policy review evidence

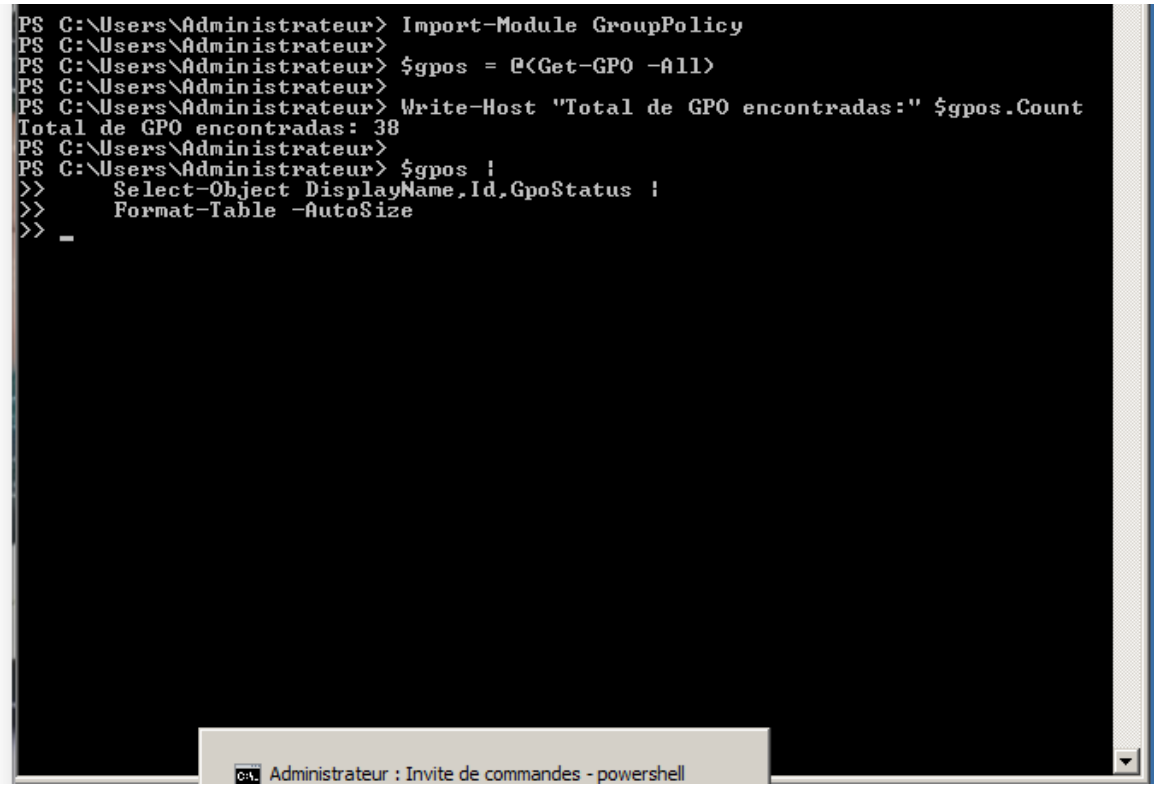


Figure 11 — Group Policy migration evidence (screenshot ad-50).

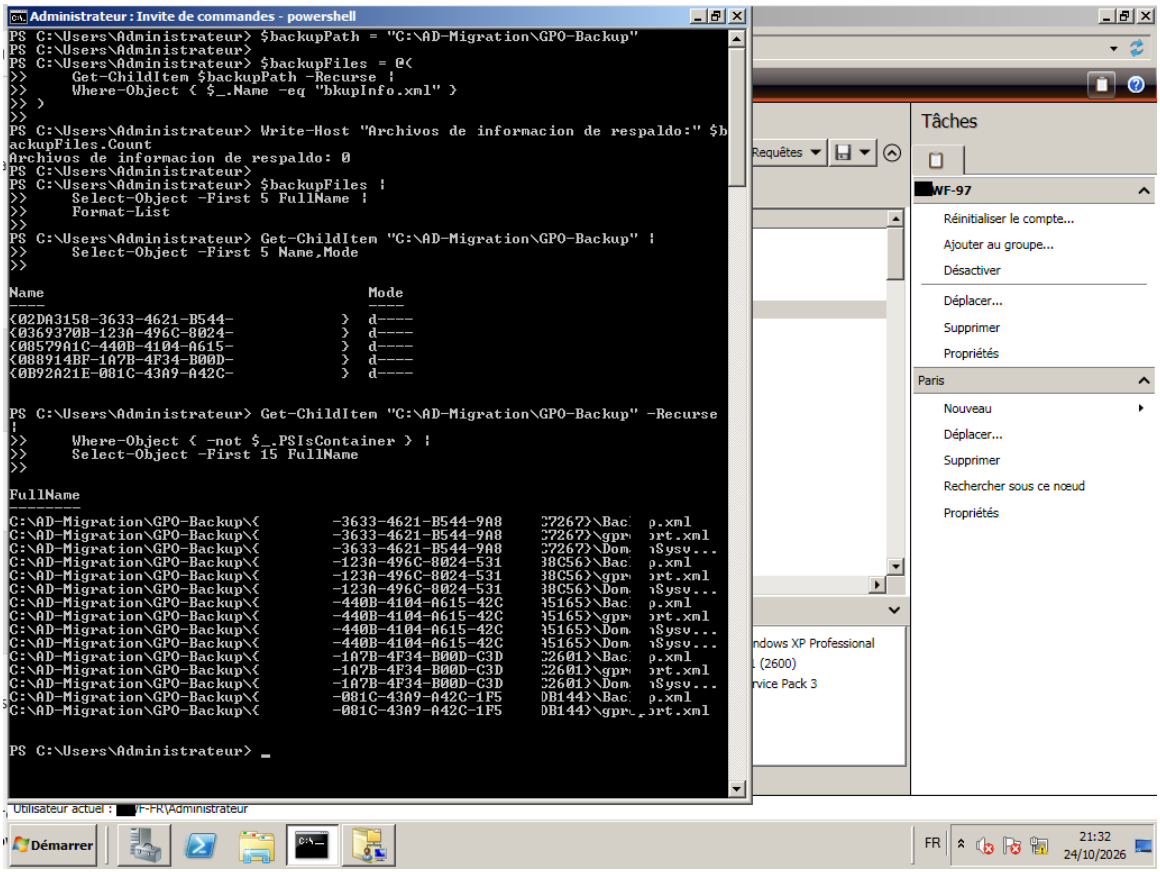


Figure 12 — Group Policy review (screenshot ad-57).

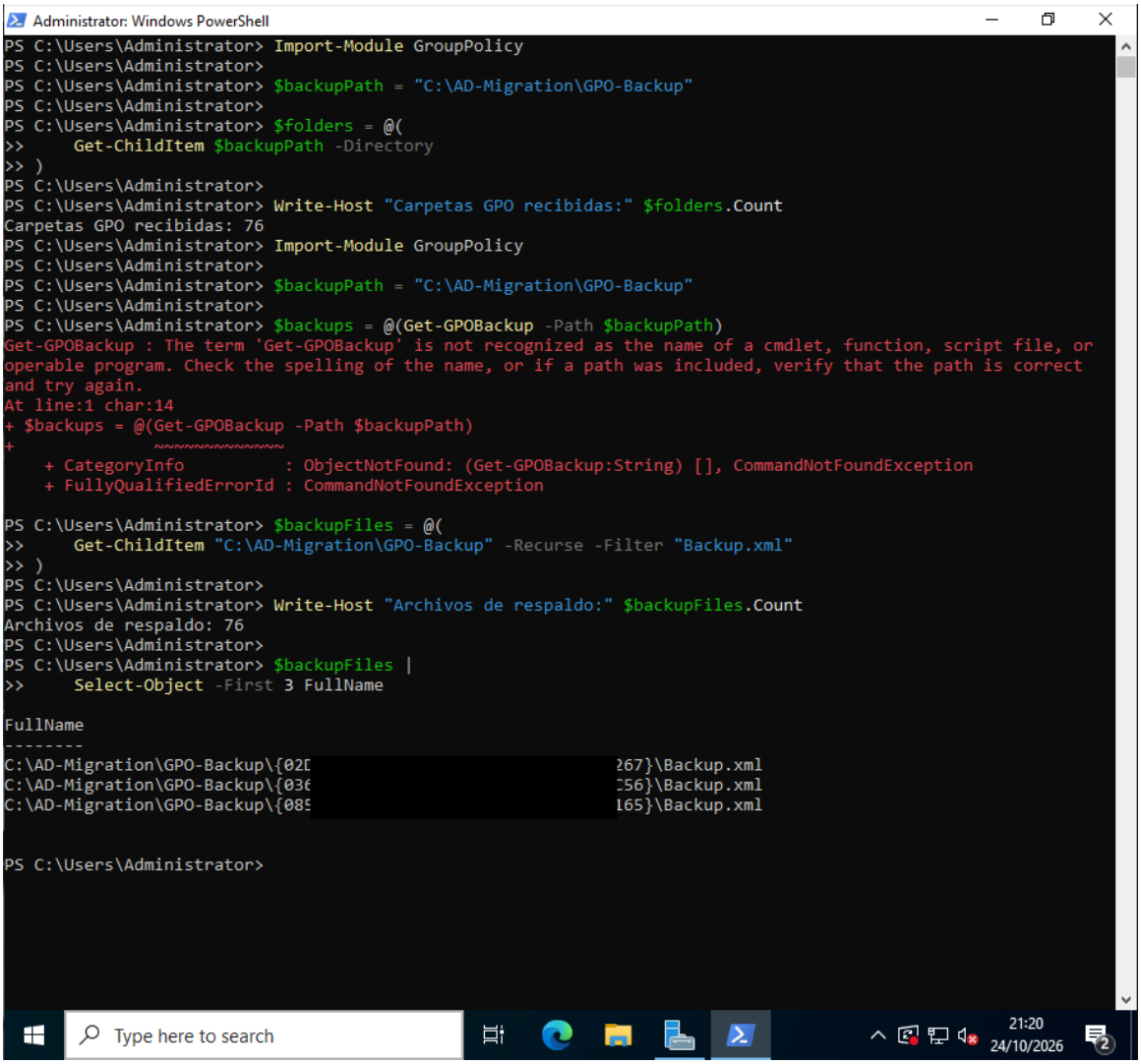


Figure 13 — Imported policy inspection (screenshot ad-64).

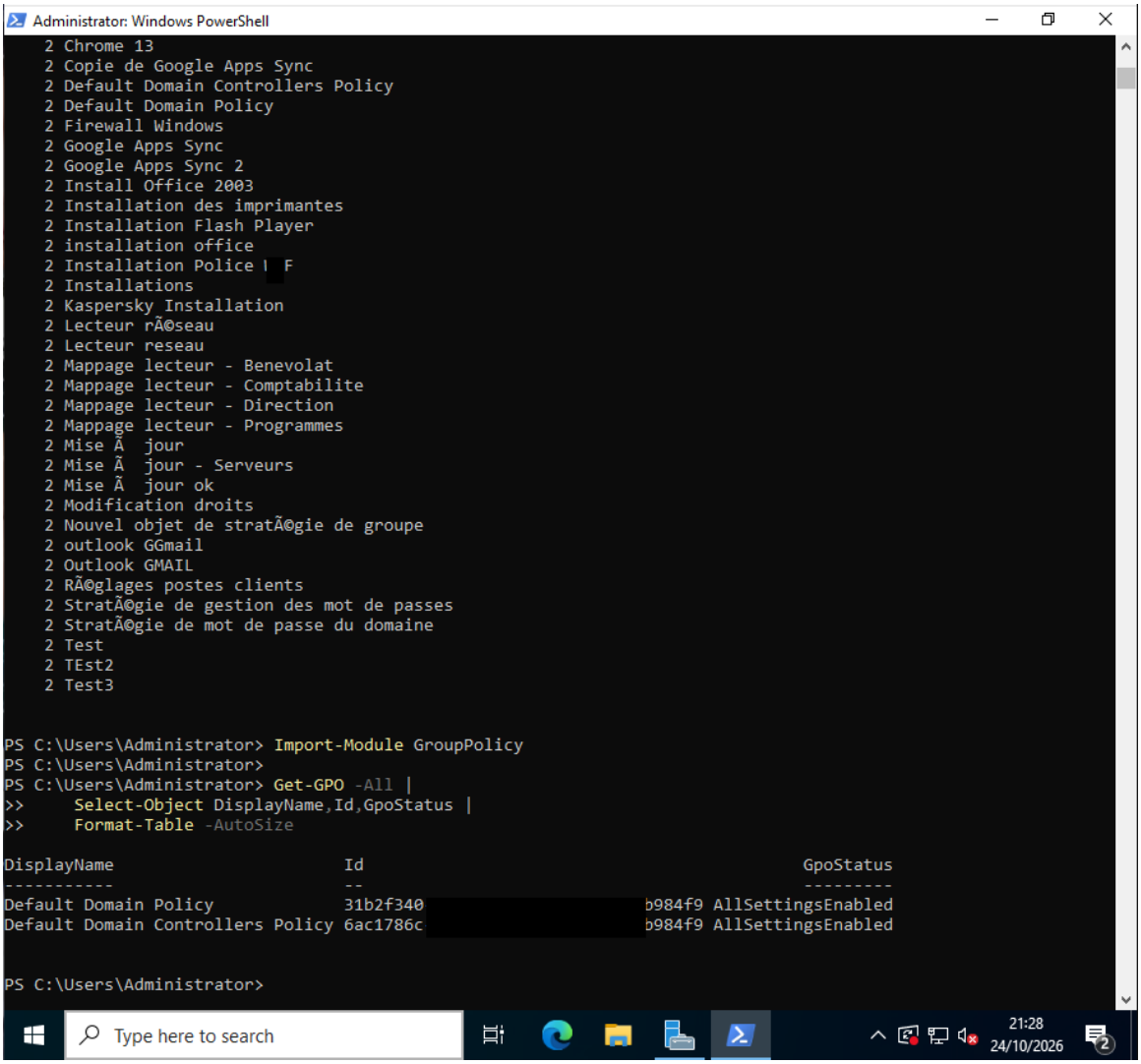


Figure 14 — Legacy policy assessment (screenshot ad-67).

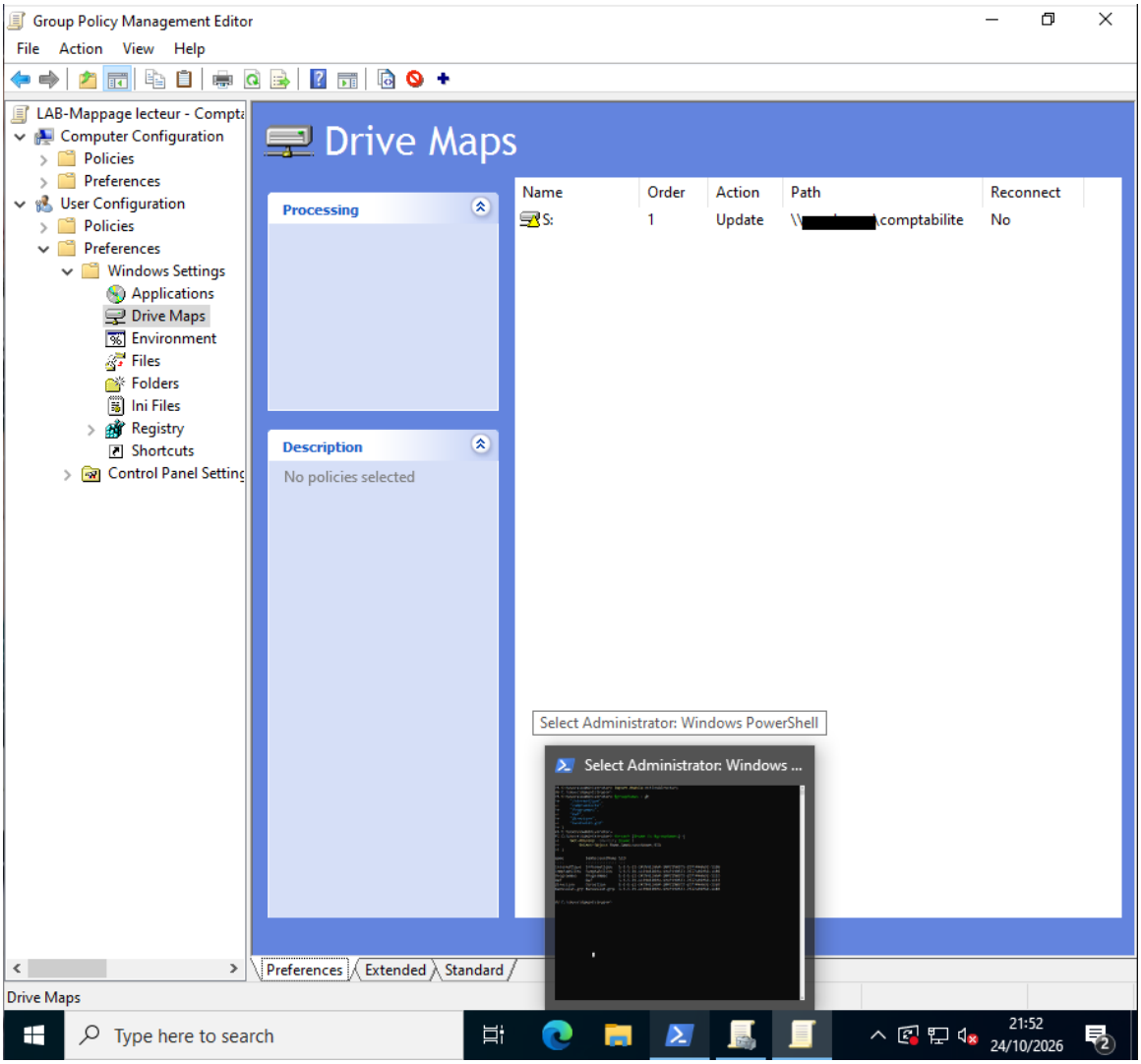


Figure 15 — Group Policy configuration review (screenshot ad-70).

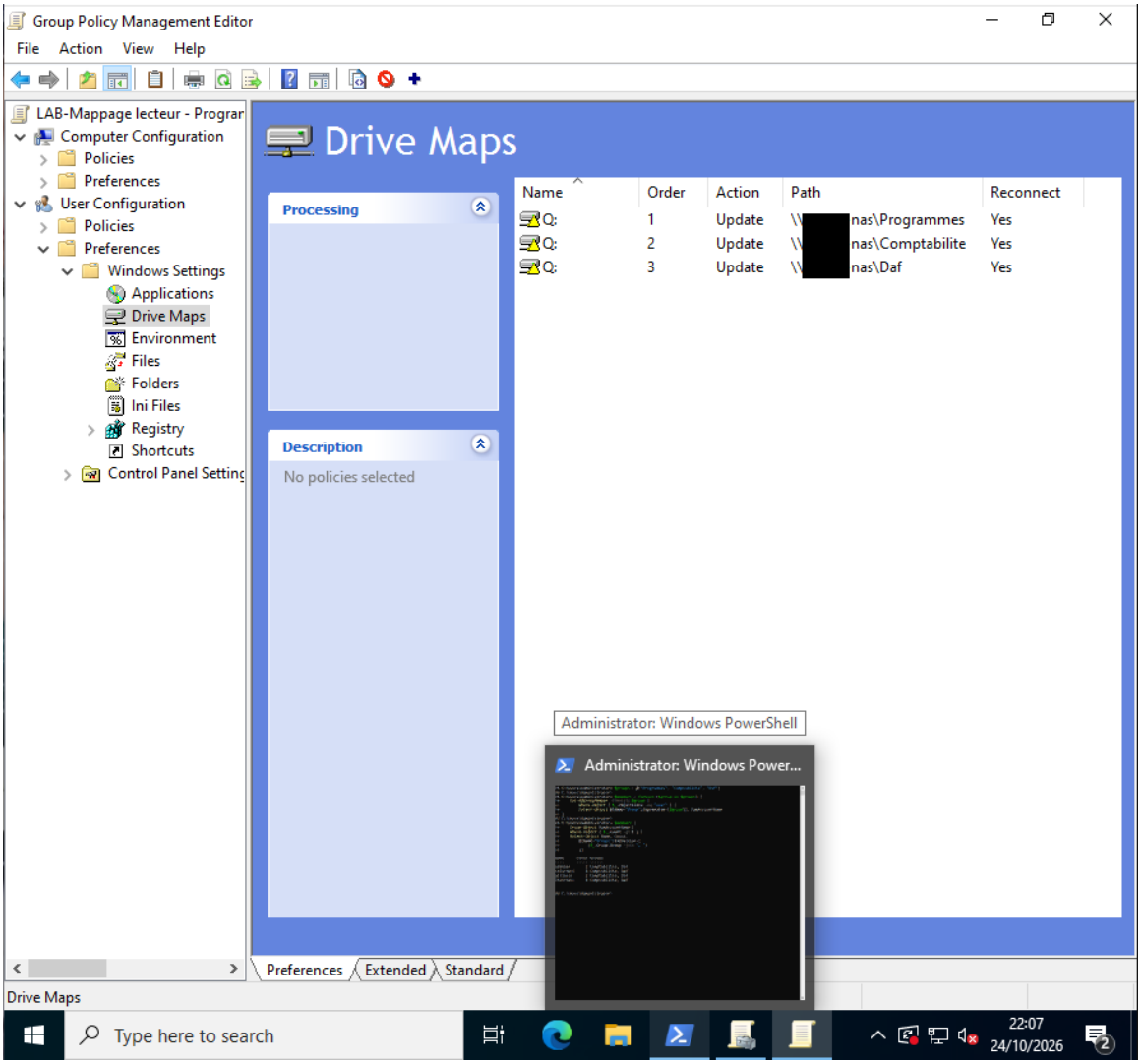


Figure 16 — Legacy settings assessment (screenshot ad-73).

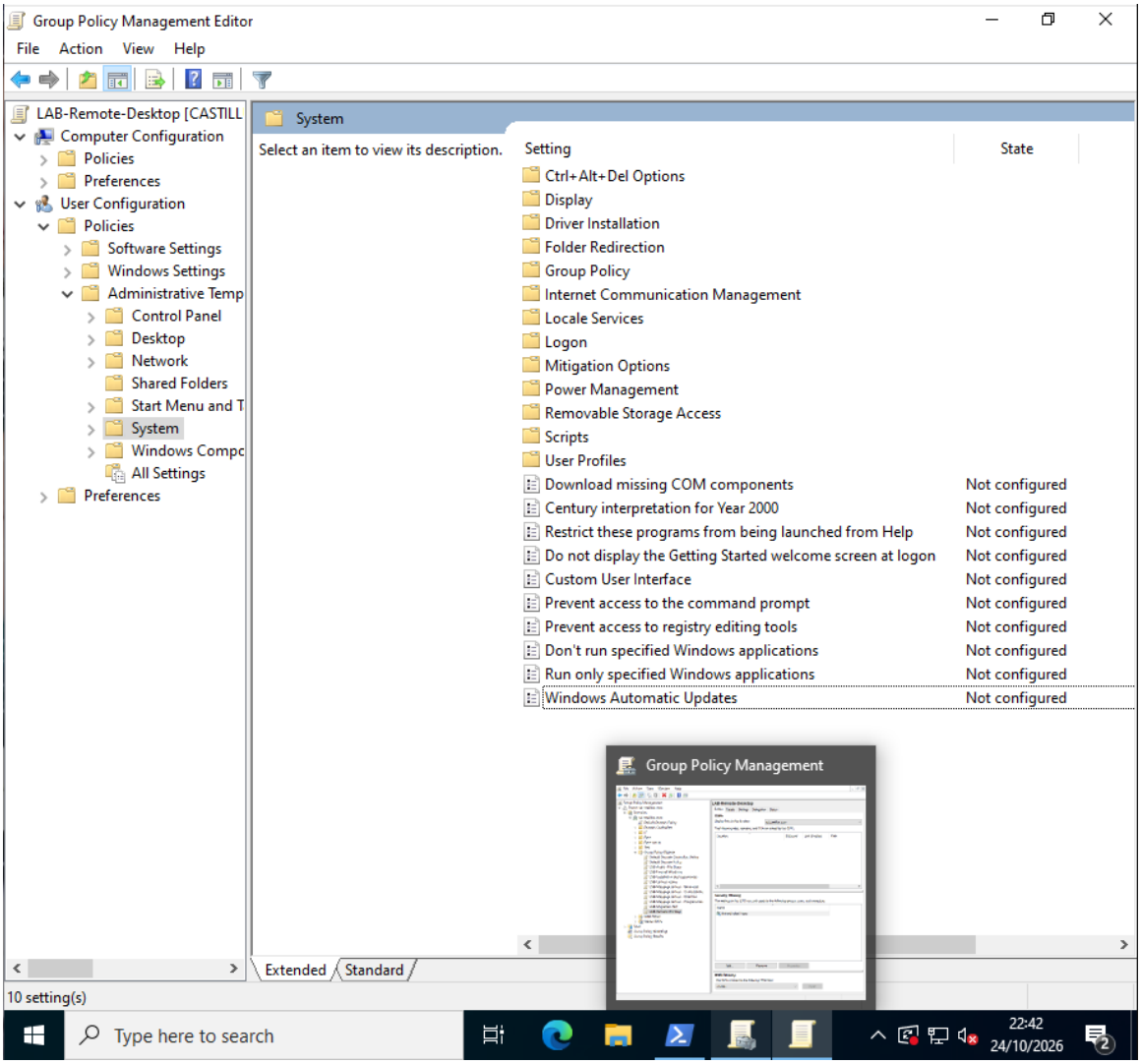


Figure 17 — Group Policy validation (screenshot ad-79).

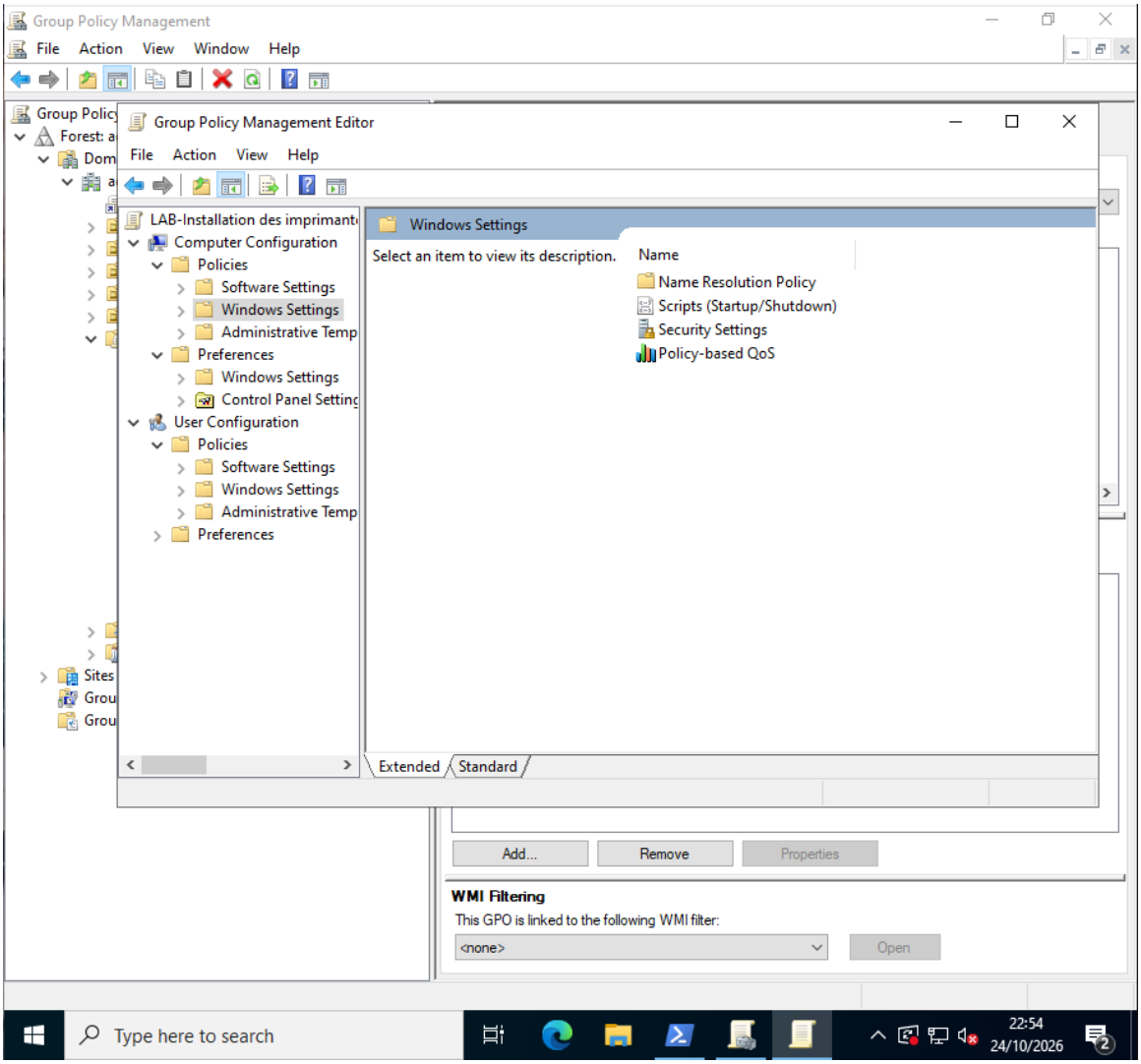


Figure 18 — Imported policy configuration (screenshot ad-83).

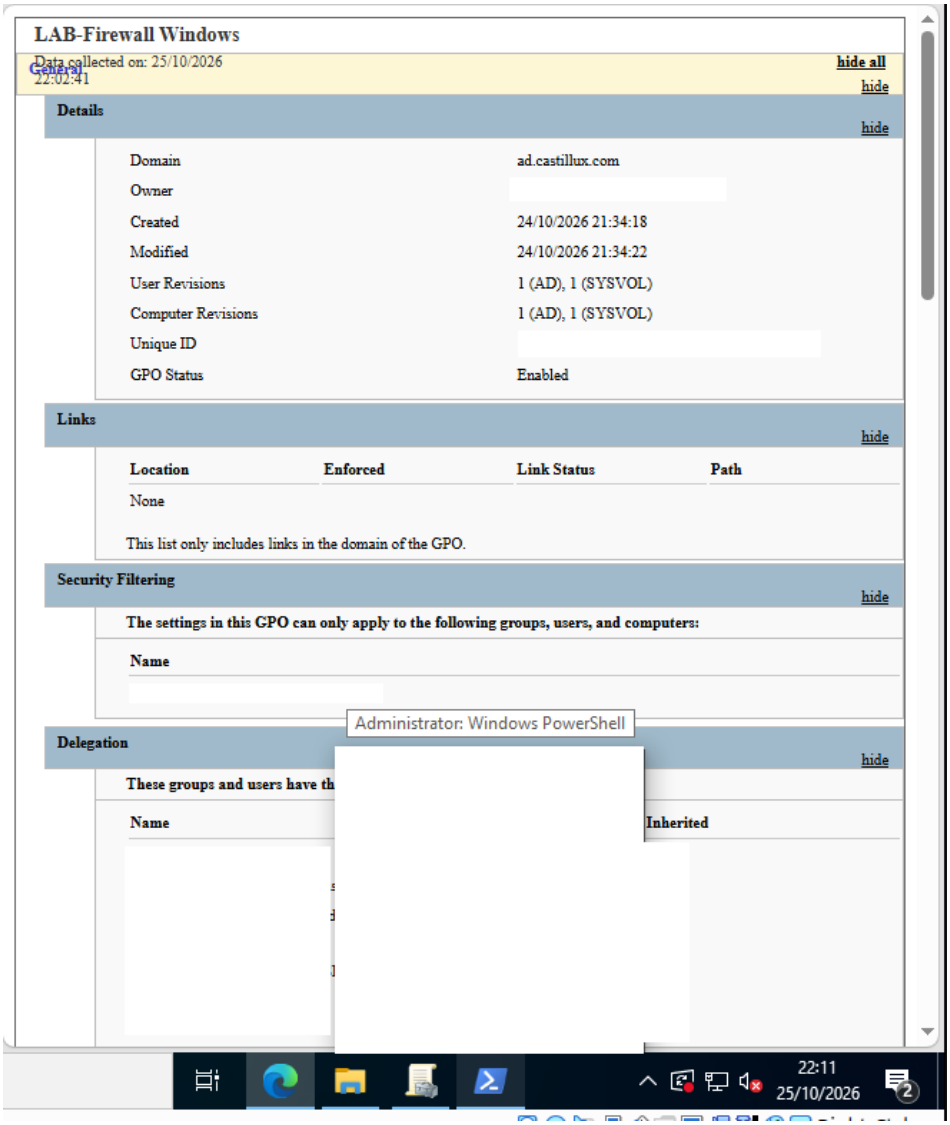


Figure 19 — Windows Firewall policy report (screenshot ad-85).

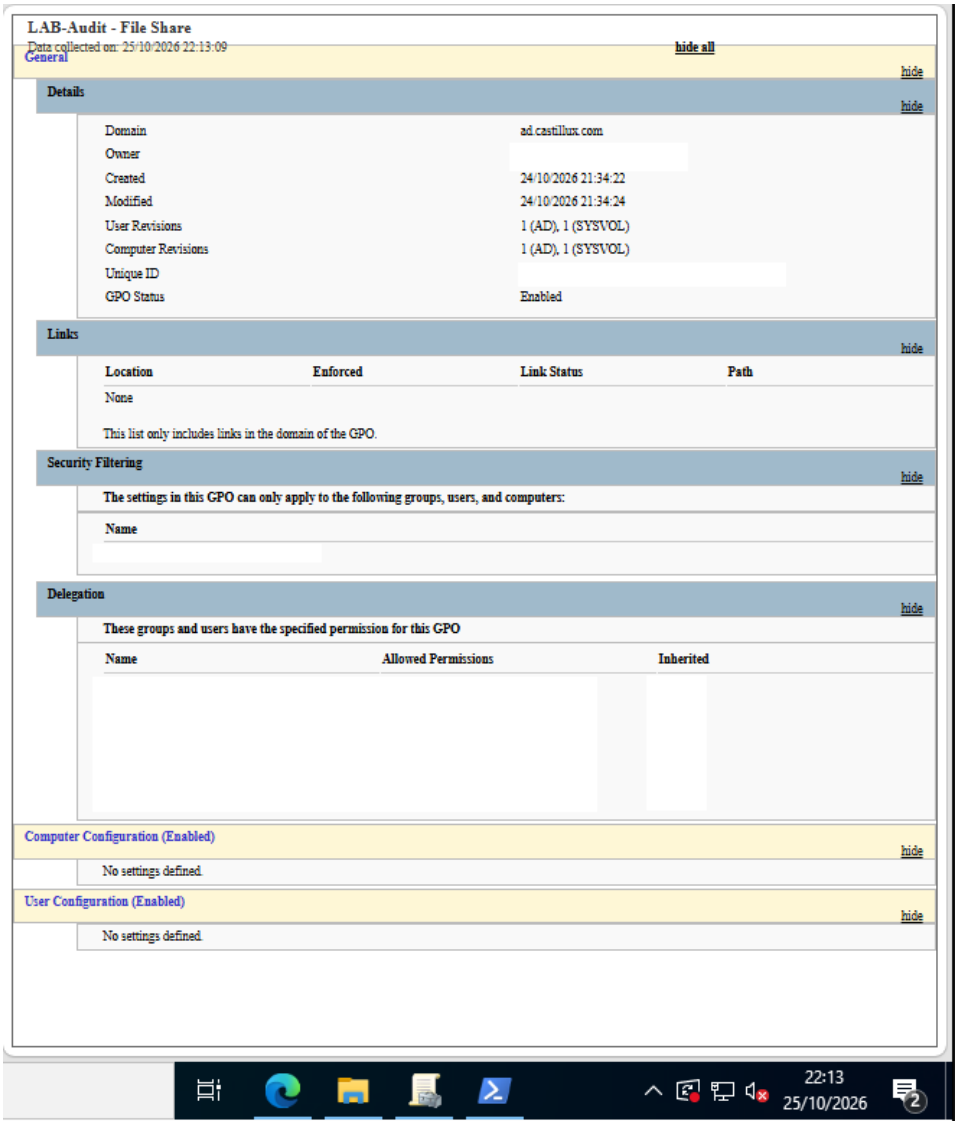


Figure 20 — File Share Audit policy report (screenshot ad-90).

6. Validation and troubleshooting

DCDIAG passed connectivity, advertising, services and DNS checks. Both SYSVOL and NETLOGON shares were published on the destination controller. These results establish core controller health, not an end-to-end production cutover.

dcdiag /test:Advertising /test:Services /test:DNS
Get-SmbShare -Name SYSVOL,NETLOGON | Select-Object Name,Path,Description

Issue
Resolution / learning
Computer import: 222 errors
Preloaded existing computer names and used -contains rather than failing identity lookup.
French GPO name encoding
Read exact display name from manifest.xml.
Get-GPOBackup not recognised
Used backup manifest and Import-GPO.
Drive Q: overlapping group targeting
Reviewed memberships and prioritised Daf.

Issue
Resolution / learning
Remote Assistance policy location
Found under Computer Configuration.
Legacy printer deployment
Inspected GPO XML and removed obsolete script reference.
Old firewall and VNC exceptions
Kept policy unlinked; recommended least-privilege replacement.

Final validation evidence

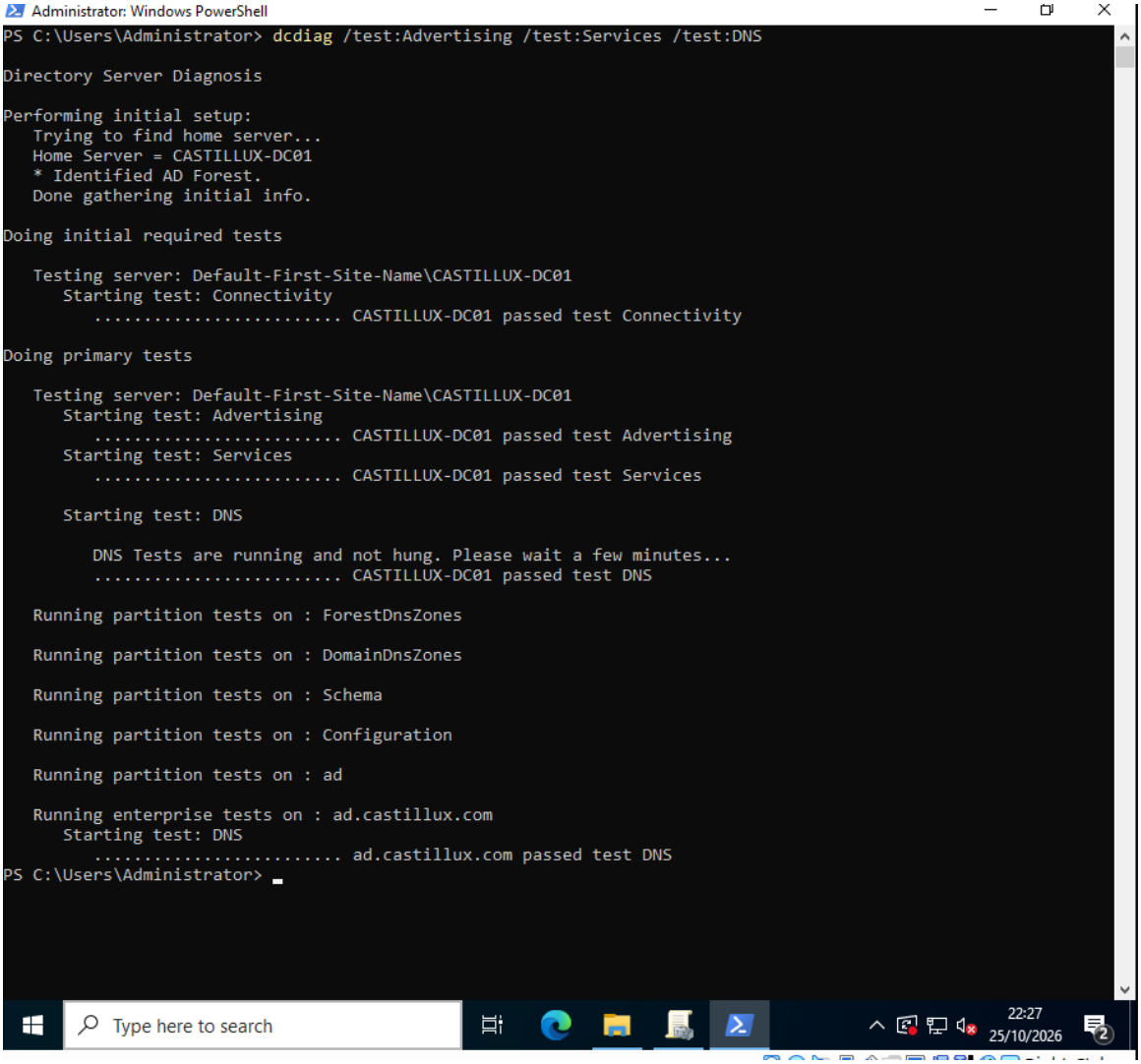


Figure 21 — Directory services validation (screenshot ad-92).

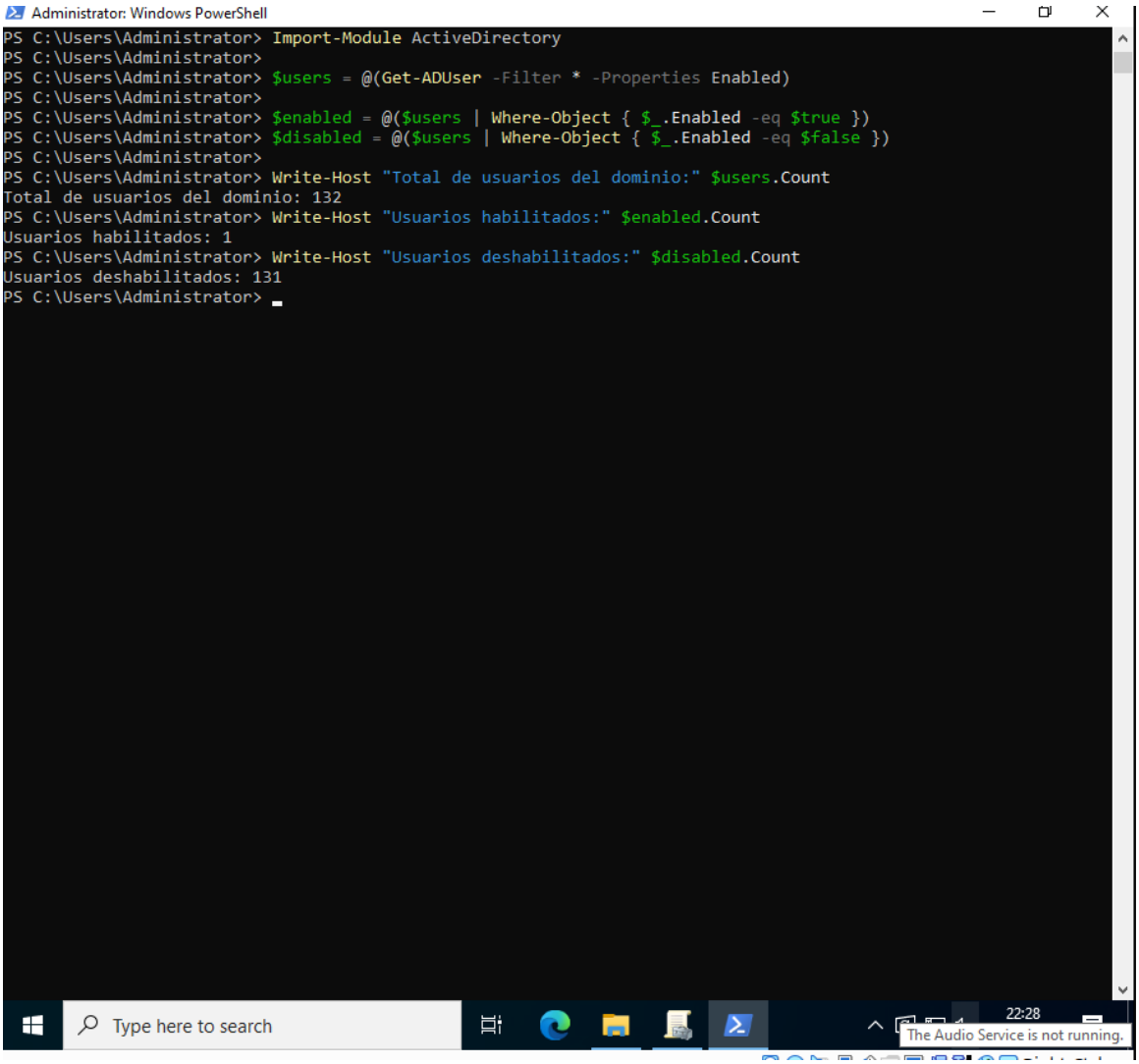


Figure 22 — Migrated account verification (screenshot ad-93).

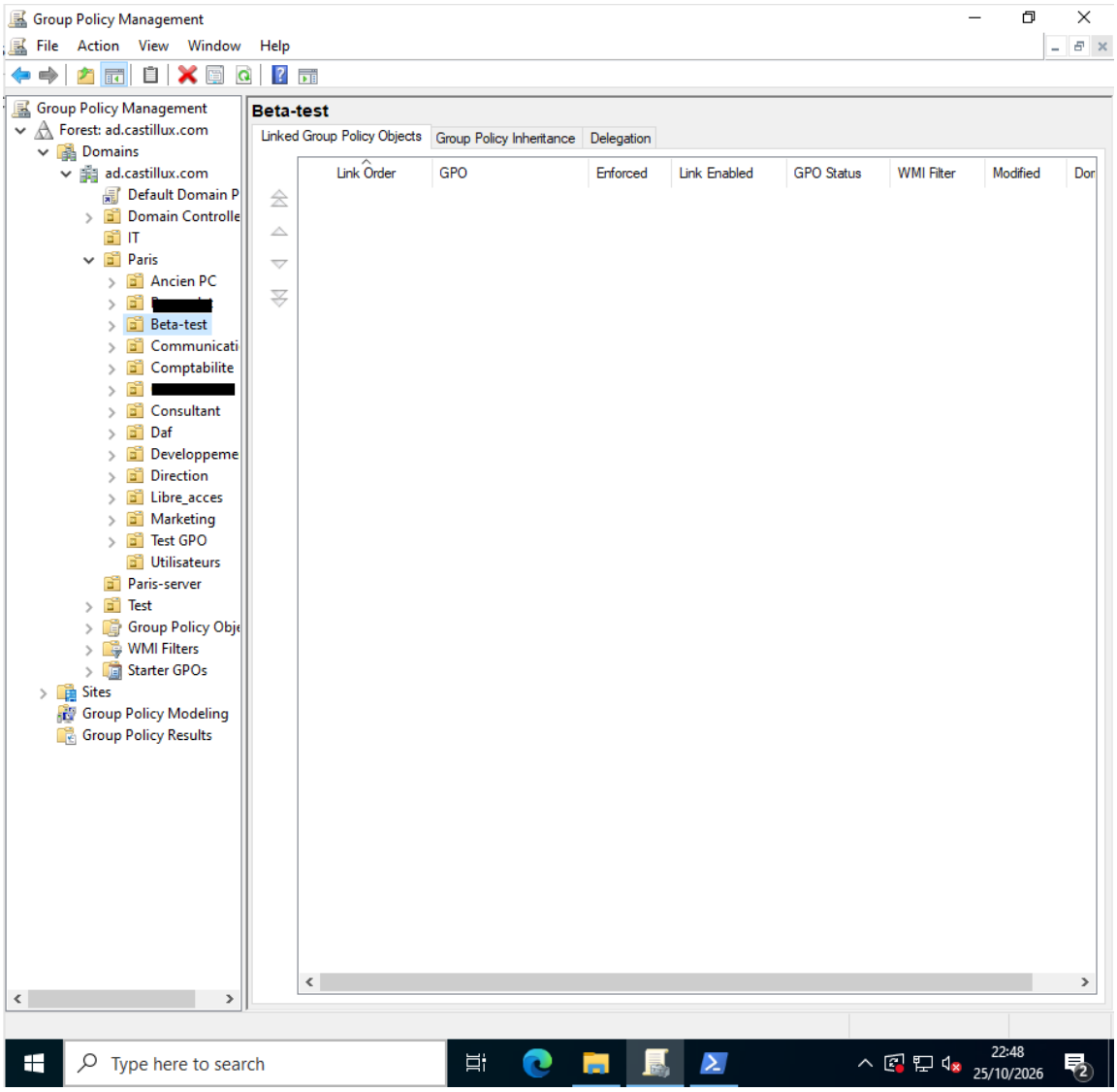


Figure 23 — Final migration verification (screenshot ad-94).

7. Completion status and next-stage work

The scoped directory-object recreation is complete: 20 custom OUs, 21 security groups, 129 disabled users, 153 direct user/group memberships and 222 disabled computer accounts. Ten historical LAB- GPOs were imported for review and left unlinked. No passwords, SIDs or machine trust relationships were migrated, and no physical computers were rejoined to the destination forest.

Recommended follow-up: independently verify the final target computer-object count; inspect migrated objects using dsa.msc; join a supported isolated Windows test client to ad.castillux.com; and design fresh least-privilege policies for a test OU. Legacy Windows XP and Windows 2000 systems require a separate compatibility and security assessment.

Appendix — Evidence and reproducibility notes

This edition includes all 23 replacement screenshots supplied in images.zip, grouped by the documented lab stages. Captions are descriptive labels and should be checked against each screenshot before public release. The source server and domain labels are public aliases; commands containing these aliases must be adapted to the actual lab if rerun. Some early export scripts are unavailable and have not been fabricated. The 222 computer accounts refer to Active Directory objects, not completed client re-joins.